Efficient Ensemble Model for Facial Expression Recognition

Team 3

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Introduction

Recent studies have developed implementations of deep learning models for **facial expression recognition** (FER) tasks because of their **potential application** in evaluating customer service satisfaction, human-computer interaction systems, and the development of accessibility features for disability-inclusive systems [6].

The **current SOTA** for image classification and FER revolves around **upscaling** the **model size** at the cost of computational resources [1], [7], [8], [10]. The challenge lies in developing models that can accurately identify emotions while maintaining computational efficiency, particularly for real-time applications.

As computational power is neither easy nor affordable to lay hands on, we aim to compare a large model's accuracy and efficiency to those of smaller models that utilize ensembling techniques. **Our vision is to find out if a way exists to achieve comparably great accuracy to that of the larger model, hence achieving great accuracy by using less computational power and without scaling the models.**

We test various ensemble methods, including majority voting, simple averaging and weighted averaging to evaluate the performance in FER tasks. The results show that a all multi-class ensemble of compact models can achieve high accuracy comparable to human performance, with significantly reduced computational demands.
This research aims to provide insights into the trade-offs between accuracy and efficiency in FER, offering practical solutions for real-world applications where computational resources are limited.

Methodology

Ensemble deep learning for facial expression recognition. The training process is done using different bootstrap samples to ensure that each network learns different features from the FER 2013 dataset. To further increase the diversity of learned features [2], each network is selected from a variety of unique compact architecture families—ShuffleNet[11], MobileNet[4], and SqueezeNet[5]—that have different approaches for extracting features efficiently from the image, as shown in Figure 3.

Furthermore, we explored using one vs rest binary classifiers as the ensemble member candidate, to encourage complementary interaction between the ensemble member. This way, some member could focus on learning difficult facial expressions and others can focus on other expressions.

Individual predictions from each model are then combined using some aggregation function, namely majority voting, Simple averaging and weight averaging, as shown in Figure 1 and 2. Additionally, we use EfficientNet [10] B3 as the larger model to compare with our ensemble method.

We explored 2 ensemble model with different members:

- Ensemble A (All multi-class classifiers): ShuffleNet, MobileNet, and SqueezeNet
- Ensemble B (With binary classifiers): MobileNet (multi), ShuffleNet (Fear), ShuffleNet (Disgust), and MobileNet (Sad)

Ensemble B clarification: We chose to predict disgust, fear, and anger separately due to their consistently low recall in individual models. For each expression, we trained the architecture with the highest recall as a binary classifier, which improved prediction accuracy. For instance, as shown in Figure 4, ShuffleNet correctly identified fear by focusing on the eyes, while MobileNet misclassified it by focusing on the mouth and nose. This approach enhanced the performance of our emotion recognition system.

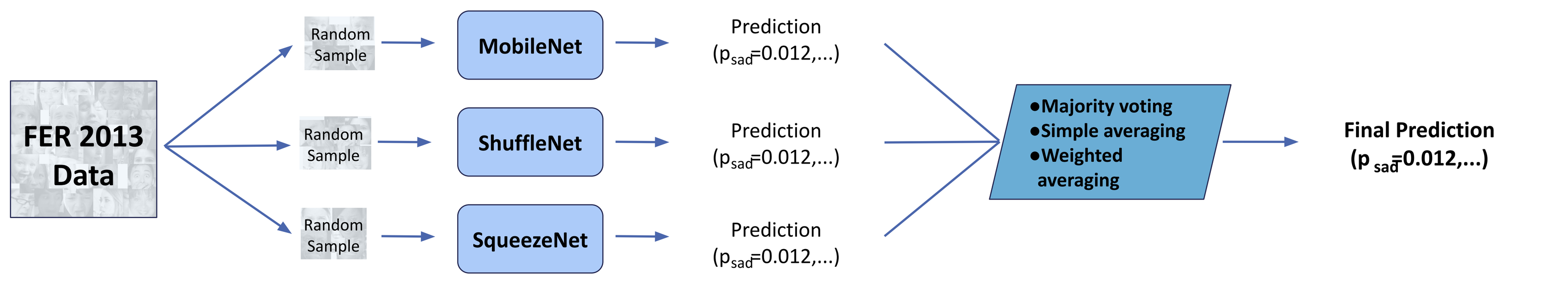

Fig 1. Ensemble A architecture

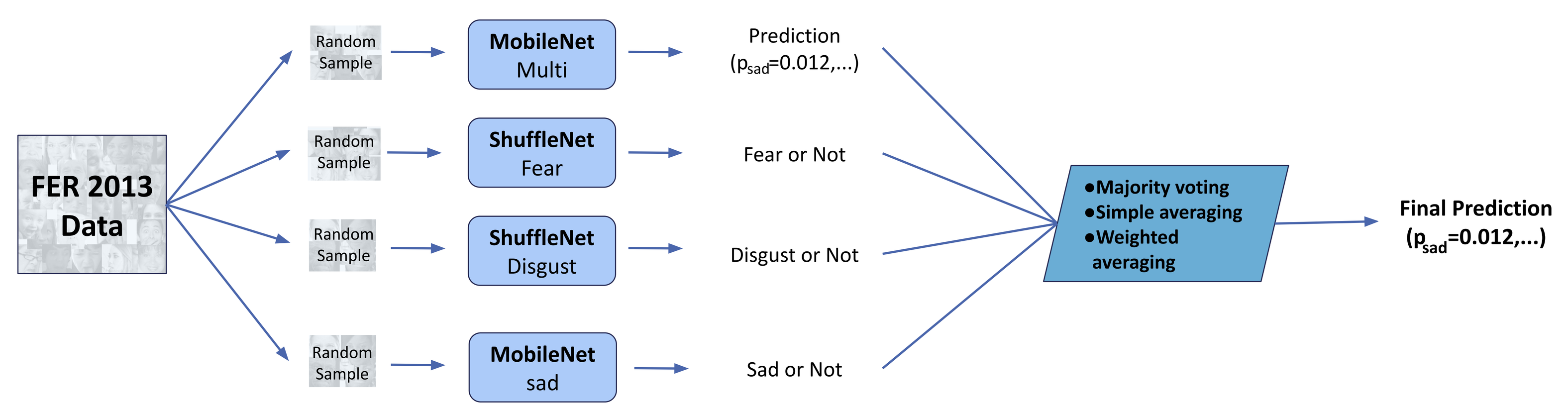

Fig 2. Ensemble B Architecture with binary classification.
Disgust, fear, and anger were chosen for binary classification due to their consistently low recall in individual models.

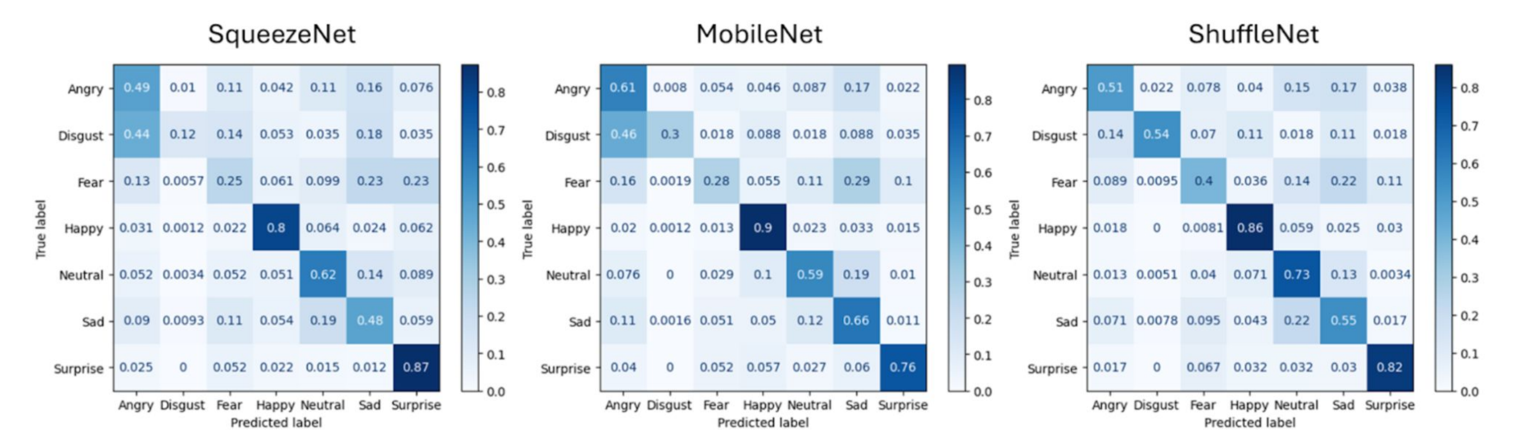

Figure 3. Confusion Matrix of ShuffleNet, MobileNet, SqueezeNet

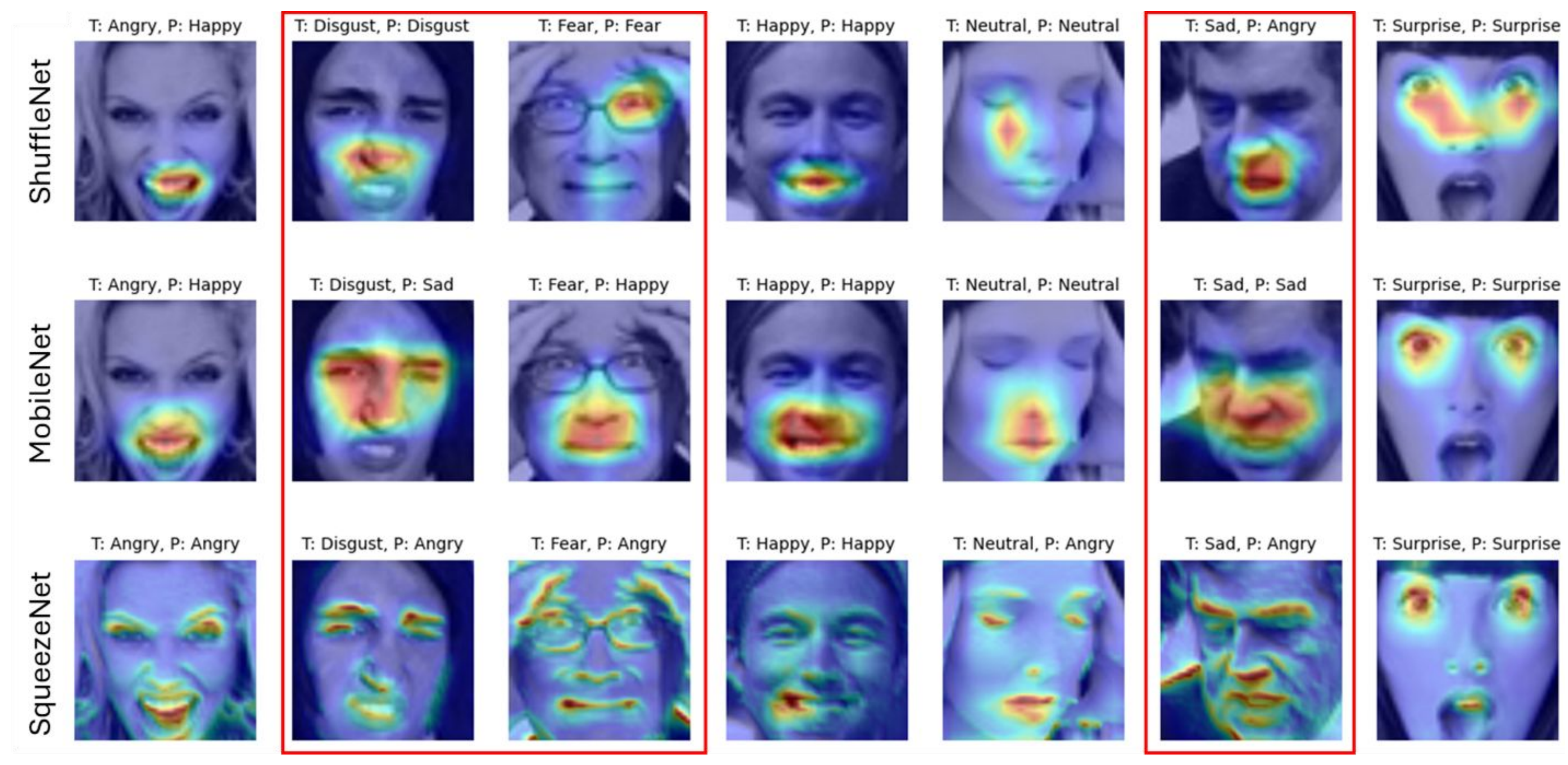

Figure 4. Grad-Cam Visualizations for ShuffleNet, MobileNet, and SqueezeNet

Results

Table 1. Performance of Specialized Ensemble Combination.

Model	Aggregation	Accuracy	Training Time	Params (M)	MACs (G)
Ensemble A	Majority voting	66.56%	**35min**	**10.30**	**1.08**
	Simple averaging	**67.01%**			
	Weighted averaging	66.95%			
Ensemble B	Majority voting	60.99%	72min	19.14	1.64
	Simple averaging	60.30%			
	Weighted averaging	56.03%			
EfficientNet B3		**68.62%**	85min	10.71	1.93
Human labeling		65.00%	-	-	-

Discussion

- This finding emphasizes the practicality and efficiency of the all multi-class ensemble approach, particularly given the context of its performance relative to human-level and large-scale model's accuracy.
- The specialized ensemble approach like Ensemble B didn't perform as well because it used a more complex approach which resulted in overconfidence (higher recall but lower precision) and more computational cost.
- It is important to select combinations that can complement each other by considering the diversity of the models rather than increasing the complexity of the ensemble, as supported by the results from Ensemble A with weighted averaging.
- Although weighted averaging was less accurate than simple averaging, It demonstrated that highlighting the strengths of each model can enhance the performance of the model.

Conclusions

We proposed 2 types of ensemble networks, ensemble A utilizing multi-class ensemble only and ensemble B utilizing one-vs-rest binary classifiers. Ensemble A has shown comparable performance to that of a larger model, while using less computational cost with less than half of the training time of the large model. These findings suggest that it is possible to create a high-performance classifier for facial expression image recognition with lower computational costs using ensemble technics as an alternative to scaling. For further research, one could explore optimizing ensemble methods to achieve even greater tradeoff between efficiency and accuracy on facial expression image recognition tasks.

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